

Machine Learning

Training Linear Regression- Week 5

Objectives

- Gradient Descent
- Linear Regression for multiple features
- Reading Assignment (Andrew Ng , CS229 Lecture notes)

Slide credits:

Andrew Ng, CS229 Lecture notes

Where to find me?

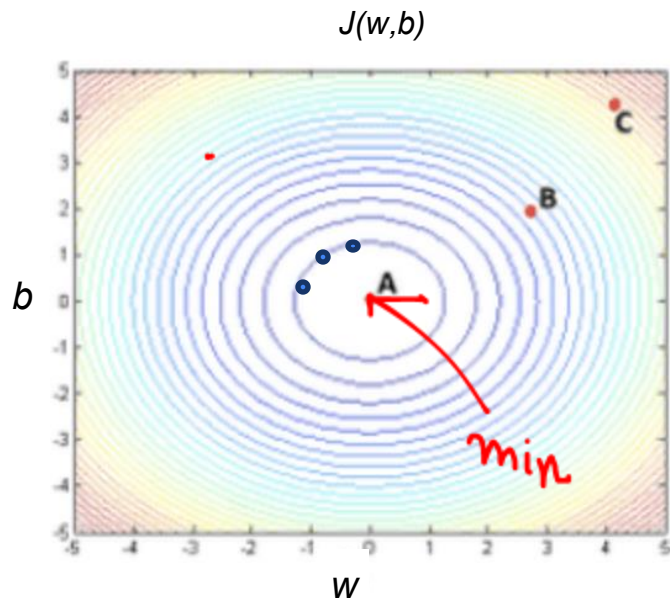
Office:

Science Hall 250

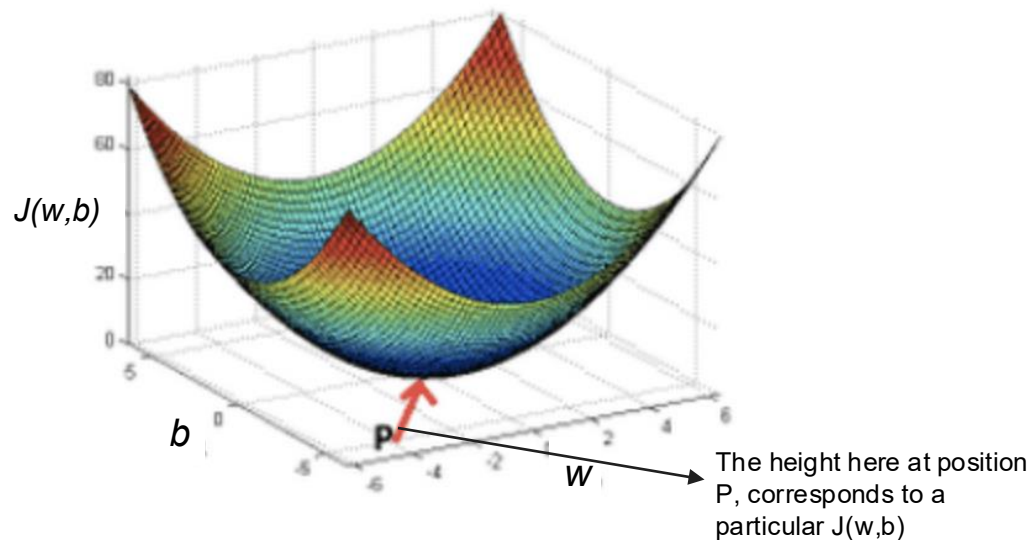
Office Hours:

**Weds and Thurs 11:00 AM - 1:00 PM EST or by
appointment**

Cost Function Visualization



Contour graph for $J(w)$



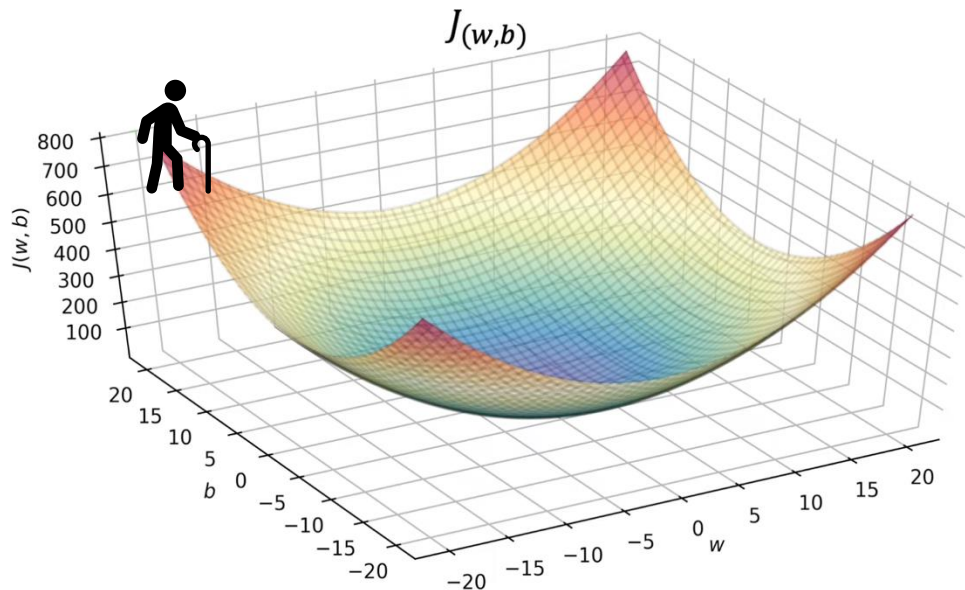
Plotting $J(w)$ for different values of w and b

Gradient Descent

The standard numerical way of minimizing $J(w, b)$ is gradient descent.

Intuitively, you can think of gradient descent as descending from a hill. We are assuming that you can't see too far, just your immediate surroundings.

You take a few steps in the direction that the hill is descending steepest. You repeat the process starting from the new point. Hopefully you will end up at the “bottom”.



Gradient Descent

The “bottom” here represents the location where $J(w, b)$ is **minimum**.

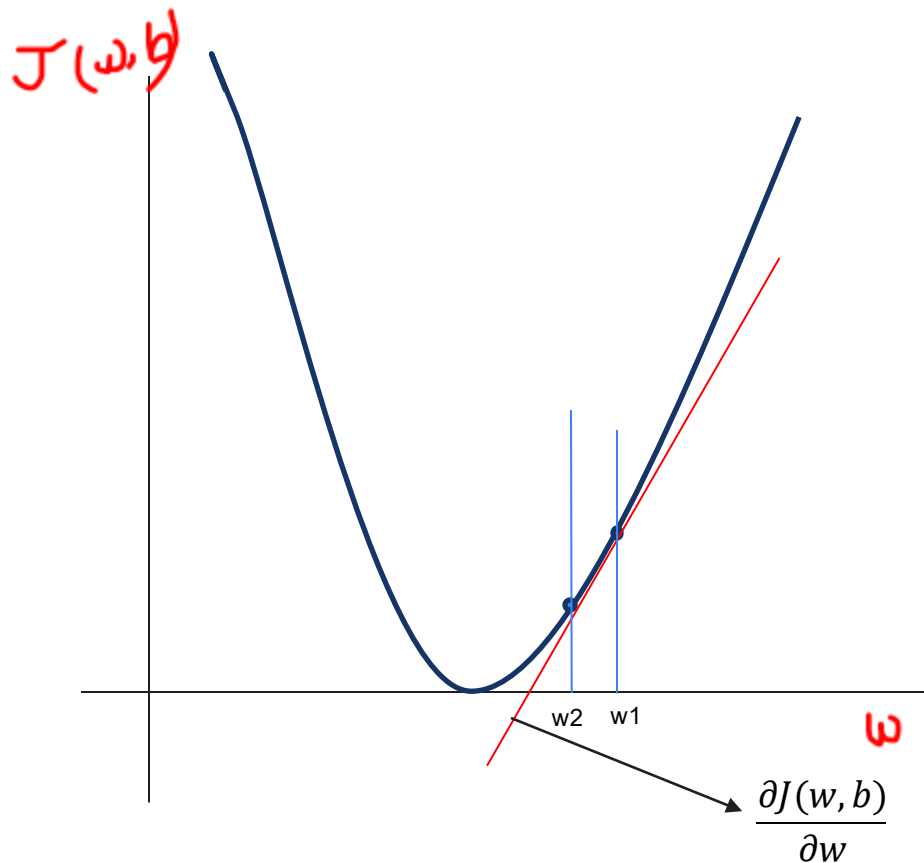
With gradient descent, we are trying to find values of w and b such that $J(w, b)$ is minimum.

Therefore, the **algorithm** can be written as:

Repeat until **convergence**:

$$w = w - \alpha \frac{\partial J(w, b)}{\partial w}$$

$$b = b - \alpha \frac{\partial J(w, b)}{\partial b}$$



Gradient Descent

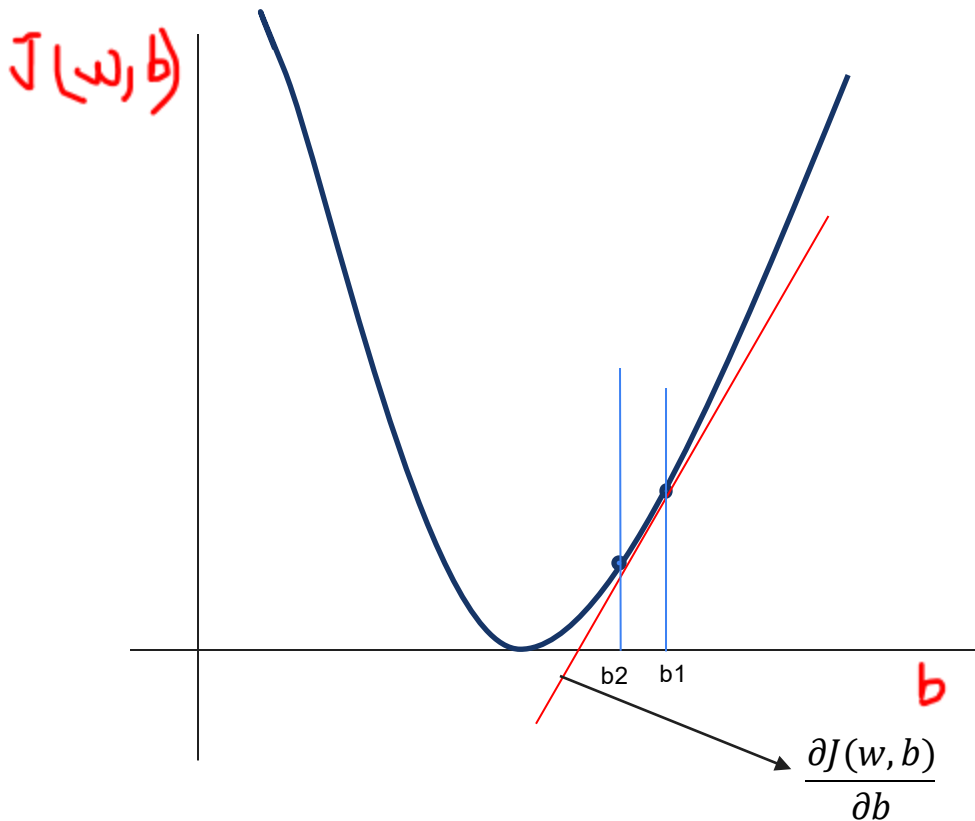
Repeat until convergence:

$$w = w - \alpha \frac{\partial J(w,b)}{\partial w}$$

$$b = b - \alpha \frac{\partial J(w,b)}{\partial b}$$

α is called learning rate. It is equivalent to the step one would take to descend the hill.

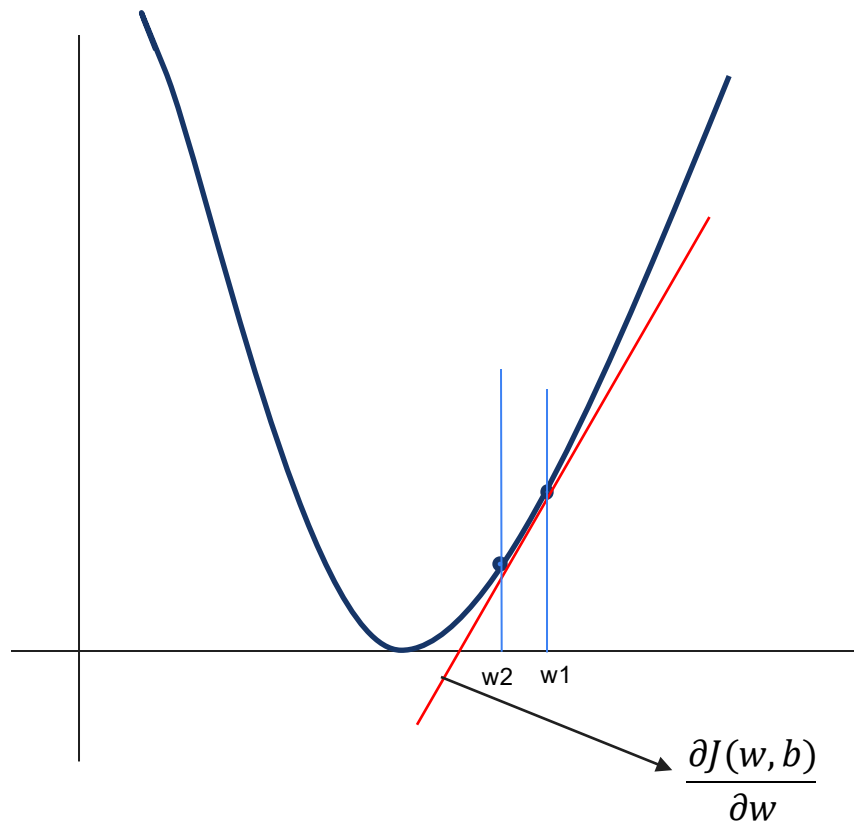
$\frac{\partial J(w,b)}{\partial w}$ and $\frac{\partial J(w,b)}{\partial b}$ provide the direction of movement



Gradient Descent

$$w_2 = w_1 - \alpha \boxed{\frac{\partial J(w,b)}{\partial w}} \rightarrow \text{positive}$$

Here $\frac{\partial J(w,b)}{\partial w}$ positive, therefore the new w_2 will be moving closer towards the **global** minimum.



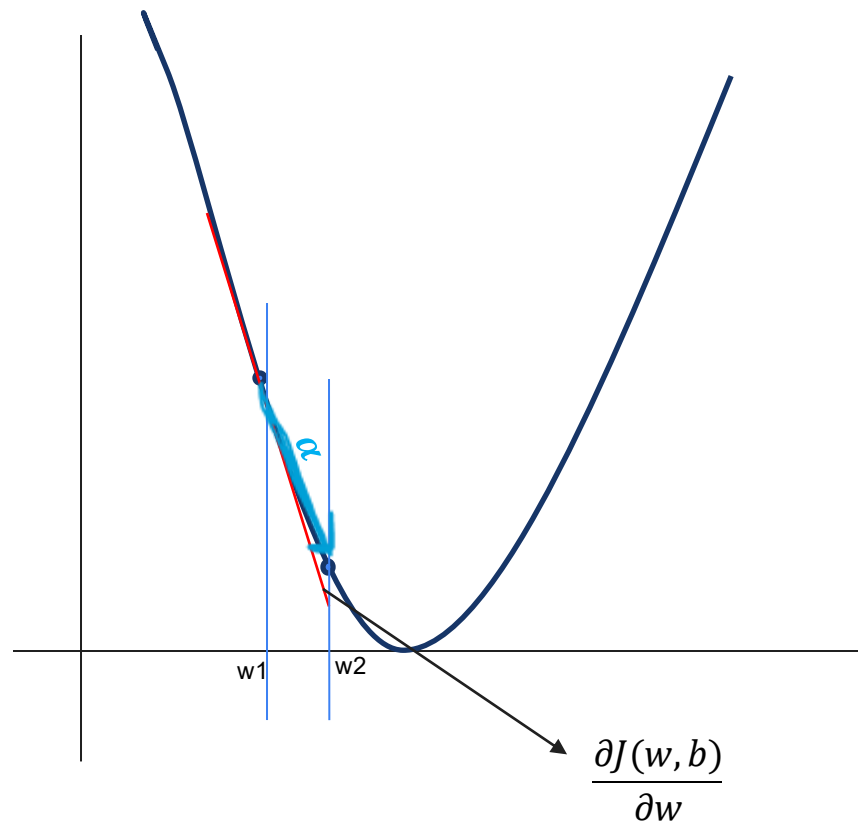
Gradient Descent

$$w_2 = w_1 - \alpha \frac{\partial J(w,b)}{\partial w} \rightarrow \text{negative}$$

Therefore, $w_2 = w_1 + \text{some number}$

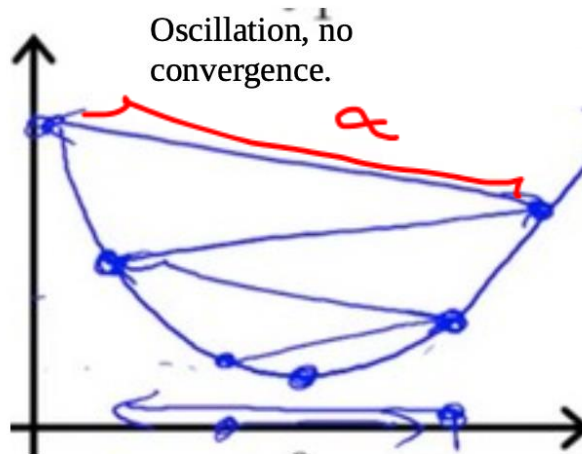
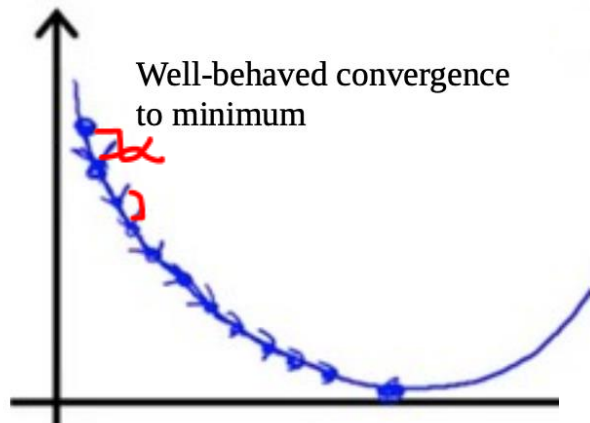
Try the visualization here:

<https://uclaacm.github.io/gradient-descent-visualiser/>



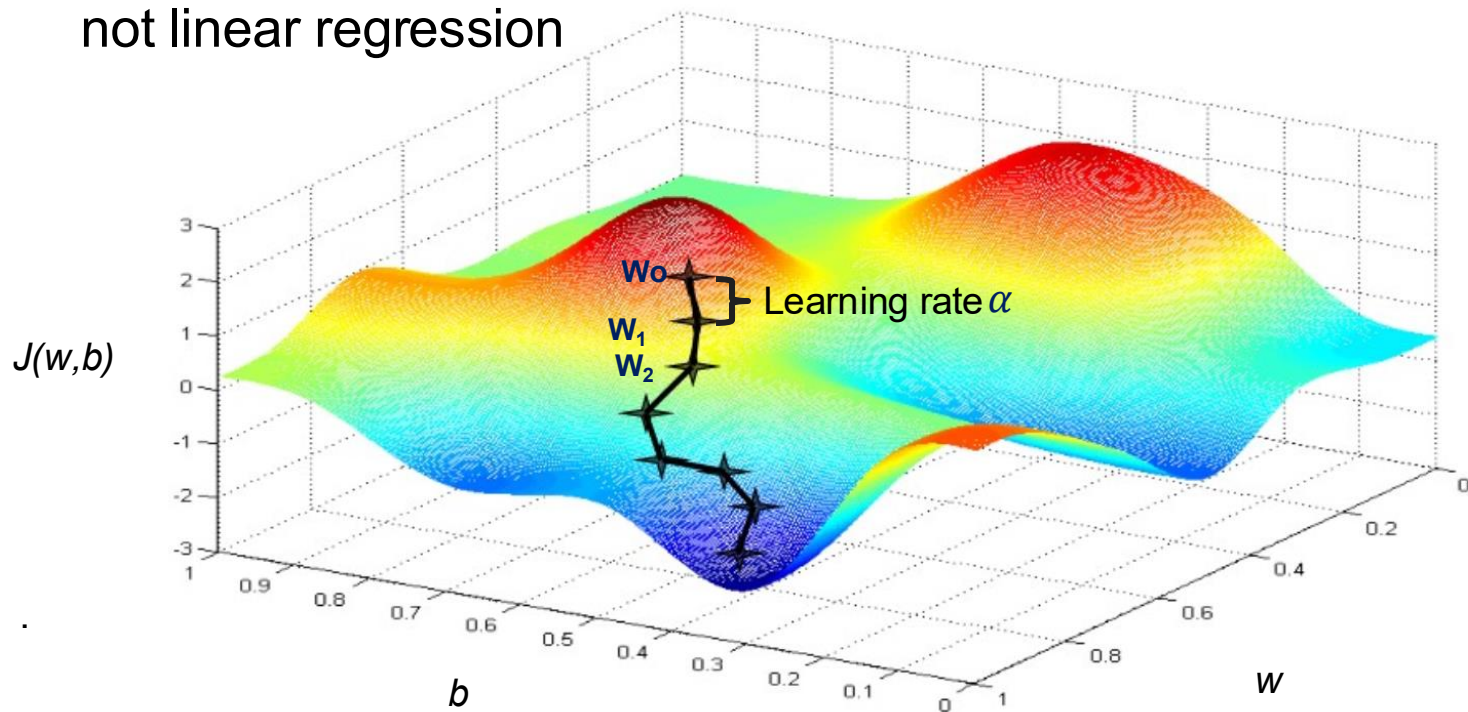
Gradient Descent- Learning Rate

- ▶ If α is too small, the rate of convergence to the minimum will be quite slow,
- ▶ If α is large, the successive iterates may oscillate wildly and you may never get to the minimum.



Gradient Descent

not linear regression



Gradient Descent

- ▶ **“Batch :”** At each training step use all the training samples to compute the parameters, w and b . Leads to stable and accurate but slow convergence.
- ▶ **“Stochastic”** At each training step, use a random data point (i.e. training sample) to calculate parameters, w and b .

Linear Regression (in Matrix Formulation)

House sizes:

2104

1416

1534

852

Predicting price of houses based on house sizes. Let's imagine, our regression line is given by the following equation

$$f(x) = -40 + 0.25 x$$

We can write training samples in the following way:

4x2 matrix

$$\begin{bmatrix} 1 & 2104 \\ 1 & 1416 \\ 1 & 1534 \\ 1 & 852 \end{bmatrix}$$

*

2x1 vector

$$\begin{bmatrix} -40 \\ 0.25 \end{bmatrix}$$

=

Predicted value (4x1)

$$\begin{bmatrix} -40 + 0.25 * 2104 \\ -40 + 0.25 * 1416 \\ -40 + 0.25 * 1534 \\ -40 + 0.25 * 852 \end{bmatrix}$$

Linear Regression (in Matrix Formulation)

House sizes: # of bedrooms Predicting price of houses based on house sizes. Let's imagine, our regression line is given by the following equation

2104 3
1416 2
1534 3
852 1

$$f(x) = -40 + 0.25x^1 + 0.3x^2$$

where $x^1 \rightarrow$ house size
 $x^2 \rightarrow$ # of bedrooms

We can write training samples in the following way:

$$\begin{bmatrix} 1 & 2104 & 3 \\ 1 & 1416 & 2 \\ 1 & 1534 & 3 \\ 1 & 852 & 1 \end{bmatrix} * \begin{bmatrix} -40 \\ 0.25 \\ 0.3 \end{bmatrix} = \begin{bmatrix} -40 + 0.25 * 2104 + 0.3 * 3 \\ -40 + 0.25 * 1416 + 0.3 * 2 \\ -40 + 0.25 * 1534 + 0.3 * 3 \\ -40 + 0.25 * 852 + 0.3 * 1 \end{bmatrix}$$

4x3 matrix 3x1 vector Predicted value (4x1)

In general, linear regression equation, can be written as : $w^T X + b$

Linear Regression (in Matrix Formulation)

In general, linear regression equation, can be written as : $w^T X + b$

Where $w^T \rightarrow$ transpose of weight matrix

$X \rightarrow$ matrix of features

Note: review linear algebra from [here](#)

ANY QUESTIONS?